

Pokhara University

Faculty of Science and Technology

Course No.: xxx xxx

Course Title: Computer Workshop (0-0-3)

Nature of the Course: Practical

Level: Undergraduate

Full Marks: 100

Pass Marks: 45

Time per Week: 3 hours in a Lab

Total Duration: 45 hours

Program: BE

Computer/IT/Software

1. Course Description

This course provides the knowledge and hands-on skills of computer hardware, software, computer networking and enables students to identify and rectify the onboard computer hardware, software and network related problems. Students will be able to understand the hardware specifications for the operating system and various application programs.

2. General Objectives

The general objectives of this course are:

- To familiarize the students with the computer systems hardware, basic practical works and computer networking concepts
- To make the students competent in install/update operating system and various application software, manage data backup and restore operations on a computer system

3. Methods of Instruction

Specific instructional techniques: Lab works, Project works

4. Contents in Detail

Specific Objectives	Contents
- Recognize the computer hardware and accessories.	Unit 1: Introduction to Desktop/Workstation Assembly 1.1 Demonstration of computer hardware and peripheral devices: Processors, Motherboard, Memory, Storage devices, I/O Devices.

	1.2 Demonstration of Power Supply-SMPS, Internal cablings and Ports.
- Assemble and Disassemble PC with power supply, internal cabling, Motherboard and PCI Device	Unit 2: Assembling Desktop Computer 2.1 Assembling steps and precautions. 2.2 Setting of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
- Assemble Laptop power supply, internal cabling, Motherboard and PCI Devices.	Unit 3: Assembling Laptop 3.1 Assembling steps and precautions. 3.2 Demonstration of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
- Install OS (Linux and Windows) in Desktop/Laptop	Unit 4: Laptop/Desktop OS Installation 4.1 BIOS setting, firmware types, BOOT configuration. 4.2 Hard disc partitioning. 4.3 OS installation (Linux, windows). 4.4 User account management 1. Printer, Scanner installation.
- Recognize computer networks-wired LAN and Wireless LAN configuration.	Unit 5: Networking and Internet Setup 5.1 Introduction to Computer Network, Network topologies, wired and wireless networking media. 5.2 Network cabling, cable types and connectors. 5.3 Basic wireless AP configuration.
- Install and update application software and utility software. - Scan and remove viruses from computer system.	Unit 6: Software Installation 6.1 Installation of application and utility software. 6.2 Update of firmware patches, 6.3 Virus scanning.
- Setup Server of Mail/Data/Domain/FTP. - Apply FTP/SCP Client tool to upload/download files to FTP server.	Unit 7: Server Installation and Backup 7.1 Introduction to server. 7.2 IP Address setting. 7.3 Study of different server (Email, Data, Domain, FTP).
- Use the basic troubleshooting tools and utilities.	Unit 8: Hardware and Software Troubleshooting 8.1 Basic Repair and Maintenance of Desktop/Laptop.

	8.2 Replacement of Passive Components. 8.3 Uses of basic software troubleshooting commands. 8.4 Uses of trouble shooting tools and utilities.
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5. Evaluation system and Students' Responsibilities

Evaluation System

The evaluation of a student may consist of attendance and class participation, lab reports, project works, viva, etc. The tabular presentation of the evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	xx	Practical		
		Attendance and Class Participation	20%	
		Project Work	40%	
		Report	20%	
		Quizzes /Viva	20%	
		Total		100

Student Responsibilities

Each student must secure at least 45% marks in the evaluation with 80% attendance in the class. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

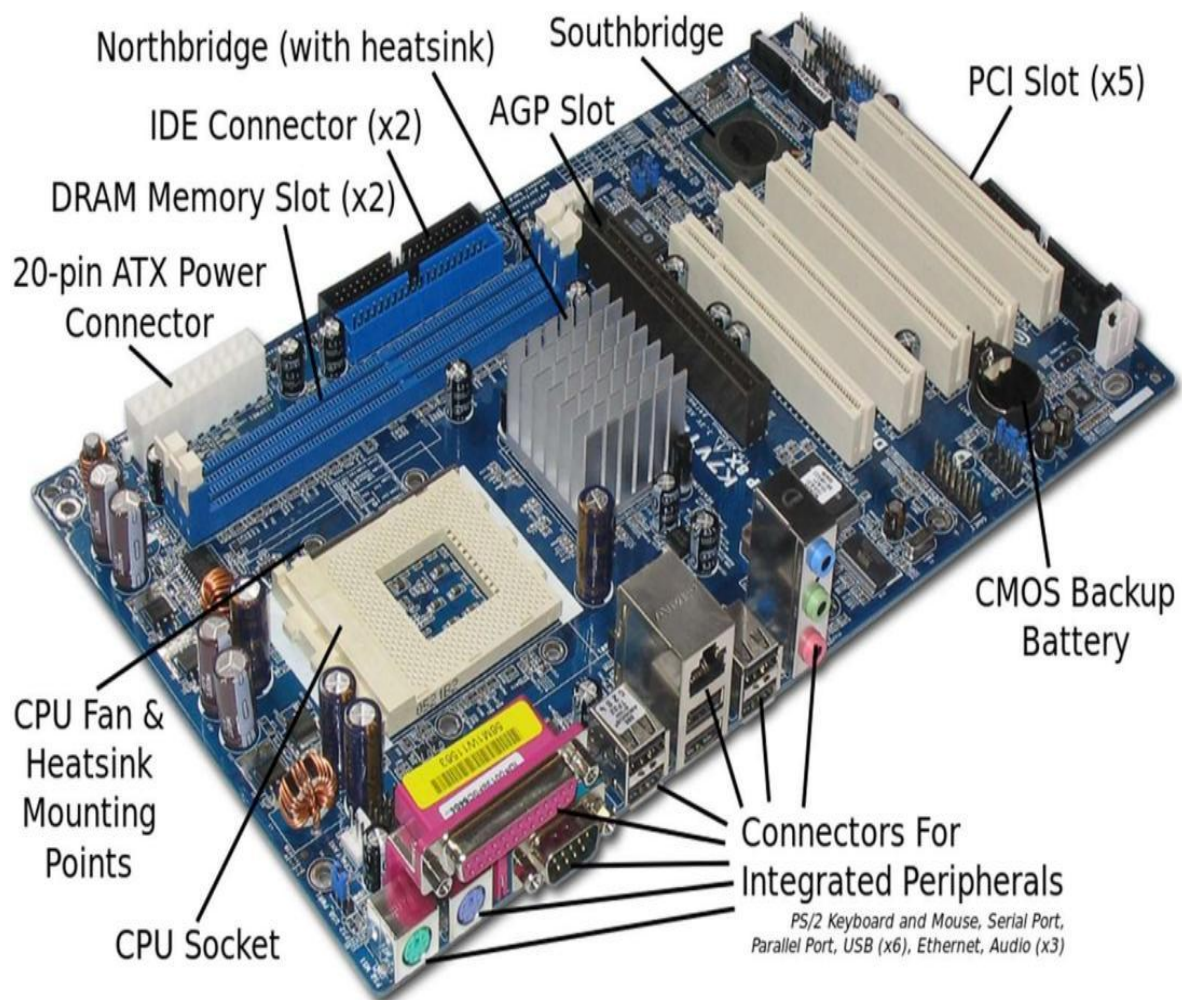
References

1. Rosch, W. R. *The Winn L. Rosch Hardware Bible*. USA: Sams.
2. Bose, A. K. *Hardware and Software of Personal Computers*. New Delhi: New Age International.
Rajaraman, V. and Adabala, N. *Fundamental of Computers*. New Delhi: PHI Learning

Lab1: Introduction To Desktop/Workstation Assembly

THEORY: Identify the peripherals of a computer, components in a CPU and its functions.

AIM: To identify the peripherals of a computer.



HARD DRIVE

Alternatively referred to as a hard disk drive and abbreviated as HD or HDD, the hard drive is the computer's main storage media device that permanently stores all data on the computer. The hard drive consists of one or more hard drive platters inside of an air sealed casing. Most computer hard drives are in an internal drive bay at the front of the computer and connect to the motherboard using either an ATA, SCSI, or SATA cable and power cable.



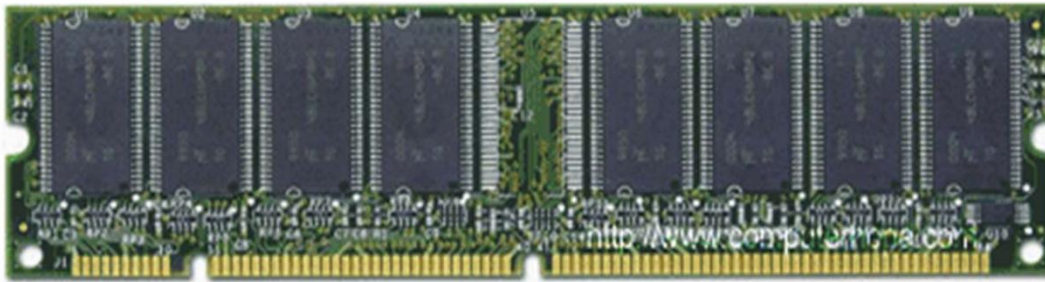
ROM

Short for **Read-Only Memory**, **ROM** is a storage medium that is used with computers and other electronic devices. As the name indicates, data stored in ROM may only be read. It is either modified with extreme difficulty or not at all. ROM is mostly used for firmware updates. A simple example of ROM is the cartridge used with video game consoles, which allows one system to run multiple games.



RAM

Alternatively referred to as main memory, primary memory, or system memory, Random Access Memory (RAM) is a computer storage location that allows information to be stored and accessed quickly from random locations within DRAM on a memory module. Because information is accessed randomly instead of sequentially like a CD or hard drive the computer can access the data much faster than it would if it was only reading the hard drive. However, unlike ROM and the hard drive RAM is a volatile memory and requires power in order to keep the data accessible, if power is lost all data contained in memory lost.



FLASH MEMORY

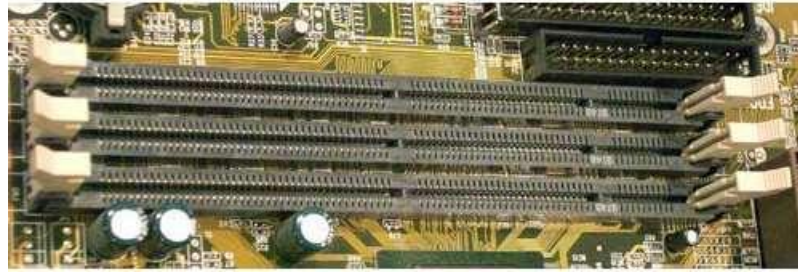
Alternatively referred to as flash storage, flash memory is a non-volatile computer memory. This type of memory is an integrated circuit that does not need continuous power to retain the data, but is a bit more expensive than magnetic storage. Today, flash memory is becoming more popular, and solid-state drives are a practical replacement for large hard drives if you have the extra money.



MEMORY SLOT

A memory slot, memory socket, or RAM slot is what allows computer memory (RAM) to be inserted into the computer. Depending on the motherboard, there may be 2 to 4 memory slots (sometimes more on high-end motherboards) and are what determine the type of RAM used with the computer.

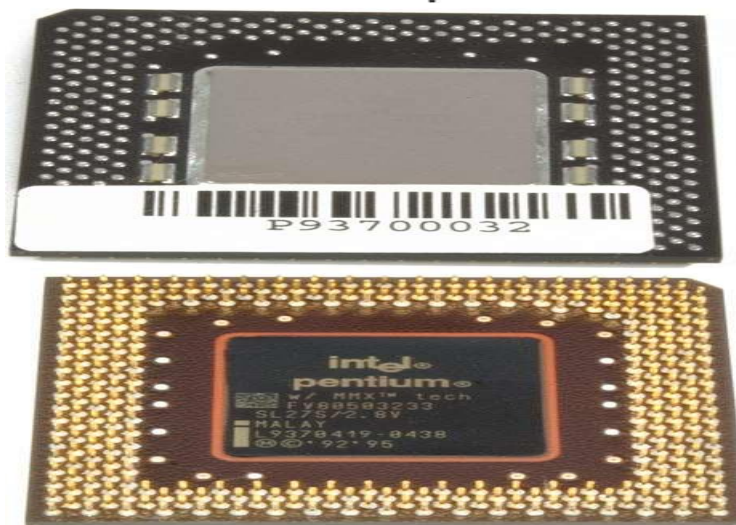
Computer Memory Slots



CPU

Alternatively referred to as the **brain of the computer**, **processor**, **central processor**, or **microprocessor**, the **CPU** (pronounced as C-P-U) is short for **Central Processing Unit**. The computer CPU is responsible for handling all instructions it receives from hardware and software running on the computer. In the picture below, is an example of what the top and bottom of an Intel Pentium processor looks like. The processor is placed and secured into a compatible CPU socket found on the motherboard, and because of the heat it produces it is covered with a heat sink to help keep it cool and running smoothly.

Intel Pentium processor



HEAT SINK

A **heat sink** is an electronic device that incorporates either a fan or a peltier device to keep a hot component such as a processor cool. There are two heat sink types: **active** and **passive**. **Passive heat sinks** are 100% reliable, as they have no mechanical components. Passive heat sinks are made of an aluminum-finned radiator that dissipates heat through convection. For Passive heat sinks to work to their full capacity, it is recommended that there is a steady air flow moving across the fins.



POWER SUPPLY

Short for **Power Supply** and sometimes abbreviated as **PSU**, which is short for **Power Supply Unit**. The **PS** is an internal hardware component used to supply the components in a computer with power by converting potentially lethal 110-115 or 220-230 volt alternating current (AC) into a steady low-voltage direct current (DC) usable by the computer.

330 Watt Power Supply (PSU)



SERIAL PORT

An Asynchronous port on the computer used to connect a serial device to the computer and capable of transmitting one bit at a time. Serial ports are typically identified on IBM compatible computers as COM (communications) ports. For example, a mouse might be connected to COM1 and a modem to COM2. With the introduction of USB, FireWire, and other faster solutions serial ports are rarely used when compared to how often they've been used in the past. The picture shows the DB9 serial port on the back of a computer.



PARALLEL PORT

Less commonly referred to as the **Centronics interface** or **Centronics connector** after the company that originally designed it, the port was later developed by Epson. The **parallel port** is found on the back of IBM compatible computers and is a 25-pin (type **DB-25**) computer interface commonly used to connect printers to the computer. Below is an example of the DB25 interface found on the back of the computer.

Parallel Port (DB25)



USB TRANSFER SPEED

USB 1.x is an external bus standard that supports data transfer rates of 12 Mbps and is capable of supporting up to 127 peripheral devices. The picture shows an example of a USB cable being connected into the USB port. USB 2.0, also known as hi-speed USB, was developed by Compaq, Hewlett Packard, Intel, Lucent, Microsoft, NEC and Phillips and was introduced in 2001. Hi-speed USB is capable of supporting a transfer rate of up to 480 megabits per second (Mbps), or 60 megabytes per second (MBps). High speed USB is backward compatible, meaning it is capable of supporting USB 1.0 and 1.1 devices and cables.

USB cable and port



Viva Questions:

- 1) Differentiate the following: Dual Core, Core 2Duo, i3, i5, i7.
- 2) Differentiate between DDR2, DDR3, DDR4
- 3) What is meant by Firmware?
- 4) Write down any 15 Keyboard Shortcut keys with description.
- 5) Compare HDD vs SSD.
- 6) Compare FAT, FAT32 with NTFS.
- 7) Write the memory storage specification from smallest unit

Lab 2: Assembling Desktop Computer

THEORY:- Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

AIM: To assemble and disassemble the system

Why should one learn about hardware?

1. Troubleshoot you and save time.
2. Knowing about system internals and components.
3. Very easy installation for modern hardware.
4. Install extra memory.
5. Removing components.

Safety Precautions:

1. Beware of electrostatic discharge (ESO)
2. Build computer on a hard surface, away from concepts.
3. Wear shoes and the short sleeved cotton wear.
4. Use Phillips, head screw driver.
5. Keep the components away from moisture.
6. Avoid using pressure while installing.

Material Required

- Computer case, with power supply installed
- Motherboard
- CPU
- Heat sink/fan assembly
- Thermal compound
- RAM module(s)

- Motherboard standoffs and screws
- Anti-static wrist strap and anti-static mat
- Tool kit

STEPS FOR ASSEMBLING

1. Setting the cabinet ready.
2. Preparing to fit the components.
3. Fitting the mother board.
4. Fitting the RAM, processor and cooler.
5. Installing PCI cards.
6. Fitting the hard disk and floppy drive.
7. Installing the CD ROM drives.
8. Connecting the ribbon cables.
9. Powering the drives and mother board.
10. Connecting the cables for the case front panel.
11. Final check.

Getting the Cabinet ready:

1. Check how to open the cabinet and determine where to fix the components.
2. Determine if the case has the appropriate risers installed.
- 3.

Preparing to fit the Components:

-Network adapter drive, Cables, Hard disk, CD-ROM Drive., RAM, CPU
, Heat sink / cooler / fan, Mother board

Fitting the Mother board.

1. Line up the patch on the motherboard (ps/l, USB, etc) with the appropriateholes in the block panel I/O shield of the case.
2. Check the points where you and to install
3. Install them and make the mother board sit on them and fix screws if required.

Mother board parts:

ACR slot., PCI Slot., AGP Slot., ATX Connectors, CPU Fan, Chipset North Bridge, CPU socket, System memory, Chipset south bridge, Panel connector, Power supply, IDE connectors.

Fitting the processor:

1. Raise the small lever at the side of the socket.
2. Notice that there is a pin missing at one corner, determine the direction to fit in the processor.
3. You should not force the CPU. When inserting it. All pins should slidesmoothly into the socket.
4. Lock the lever back down.
5. Install the heat sink over it (Different type for each processor).
Heat sink /CPU fan.

Fitting the RAM:

1. The RAM must be suitable for motherboard.
2. There are currently 3 types of RAM available.
 - a) SD RAM.
 - b) DDR SD RAM.
 - c) RD RAM.

The mother board's chipset determines which type of RAM may be used

Installing the PCI Cards:

1. Most of the cards are inbuilt these days.
2. NIL, Sound Cards etc. are fitted into PCI slots.

Fitting the hard disk and Floppy disk:

1. Place the floppy and hard disks in their slots.
2. Leave some space above HDD to prevent heat building.
3. Check the jumper configuration.
4. Fix the screws.

. Installing the CD-ROM Drives:

1. CD-ROM drive is similar to installing a hard disk.
2. 1ST check that the jumper configuration is correct.
3. Fix the screw.

Connecting the Cables:-

1. Attach the long end of the cable to the IDEU connector on the motherboard first.
2. The red stripe on the IDE cable should be facing the CD Power.

Powering the driver and motherboard:

Connecting the cables for the case front pane

1. SD, SPK or SPEAK: The loud speakers o/p. it has 4 pins.
2. RS, RE, RS or RESET: Connect the two pin Reset cable here.
3. PWR, PW, PWSW, PS or power SW: Power switch, the pc's on (switch, the plug is two pin).
4. PWLED, PWRLED or Power LED: The light emitting diode on the front panel of the case illuminates when the computer is switched on. It's a 2-pin cable.
5. HD, HDD, and LED: These two pins connect to the cable for the hard disk activity LED.

Final Check:-

1. Mother board jumper configurations are the settings for the processor operator.
2. Drive jumper settings, master/ slave correct?
3. Are the processor, RAM modules and plug in cards finally

seated in there sockets?

4. Did you plug all the cables in? Do they all fit really?
5. Have you tightened all the screws in plug- in cards or fitted the clips?
6. Are the drive secure?
7. Have u connected the power cables to all driver?

Powering up for the first time:

1. Ensure that no wires are touching the CPU heat sink fan.
2. Plug your monitor, mouse and keyboard.
3. Plug in power card and switch the power supply.
4. If everything is connected as it should be
 - All system, fans should start spinning.
 - U should hear a single beep and after about 5-10 sec.
 - Amber light on monitor should go green.
 - You will see computer start to boot with a memory check.
 - Now check front LED'S to see if u plugged them in correctly.
 - Check all other buttons.
 - Power afford change any wrong settings.

DISASSEMBLING THE COMPUTER SYSTEM

● Detach the power cable:

The disassembling of the computer system starts with externally connected device detachment. Make sure the computer system is turned off, if not then successfully shut down the system and then start detaching the external devices from the computer system. It includes removing the power cable from electricity switchboard, then remove the cable from SMPS (switch mode power supply) from the back of the CPU Cabinet. Do not start the disassembling without detaching the power cable from the computer system. Now remove the remaining external devices like keyboard, mouse, monitor, printer or scanner from the back of CPU cabinet.

- **Remove the Cover:**

The standard way of removing tower cases used to be to undo the screws on the back of the case, slide the cover back about an inch and lift it off. The screwdrivers as per the type of screw are required to do the task.

- **Remove the adapter cards:**

Make sure if the card has any cables or wires that might be attached and decide if it would be easier to remove them before or after you remove the card. Remove the screw if any, that holds the card in place. Grab the card by its edges, front and back, and gently rock it lengthwise to release it.

- **Remove the drives:**

Removing drives is easier. There can be possibly three types of drives present in your computer system, Hard disk drive, CD/DVD/Blue-ray drives, floppy disk drives (almost absolute now a day). They usually have a power connector and a data cable attached from the device to a controller card or a connector on the motherboard. CD/DVD/Blue Ray drive may have an analog cable connected to the sound card for direct audio output.

The power may be attached using one of two connectors, a Molex connector or a Berg connector for the drive. The Molex connector may require to be wiggled slightly from side to side and apply gentle pressure outwards. The Berg connector may just pull out or it may have a small tab which has to be lifted with a screwdriver.

Now Pull data cables off from the drive as well as motherboard connector. The hard disk drive and CD/DVD drives have two types of data cables. IDE and SATA cables. The IDE cables need better care while being removed as it may cause the damage to drive connector pins. Gently wiggle the cable sideways and remove it. The SATA cables can be removed easily by pressing the tab and pulling the connector straight back.

- Now remove the screws and slide the drive out the back of the bay.
- Remove the memory module:

Memory modules are mounted on the motherboard as the chips that can be damaged by manual force if applied improperly. Be careful and handle the chip only by the edges. SIMMs and DIMMs are removed in a different way:

SIMM - gently push back the metal tabs while holding the SIMM chips in the socket. Tilt the SIMM chip away from the tabs until a 45° angle. It will now lift out of the socket. Put SIMM in a safe place.

DIMM- There are plastic tabs on the end of the DIMM sockets. Press the tabs down and away from the socket. The DIMM will lift slightly. Now grab it by the edges and place it safely. Do not let the chips get dust at all.

- remove processor
- remove processor
- remove the motherboard
- Remove the power supply:

The power supply is attached into tower cabinet at the top back end of the tower. Make sure the power connector is detached from the switchboard. Start removing the power connector connected to motherboard including CPU fan power connector, cabinet fan, the front panel of cabinet power buttons and all the remaining drives if not detached yet.

Now remove the screws of SMPS from the back of the cabinet and the SMPS can be detached from the tower cabinet.

- **Remove the motherboard:**

Before removing all the connectors from the motherboard, make sure u memorize the connectors for assembling the computer if required, as that may require connecting the connectors at its place. Remove the screws from the back of the motherboard and you will be able to detach it from the cabinet. Now remove the CPU fan from the motherboard. The heat sink will be visible now which can be removed by the pulling the tab upward. Finally, the processor is visible now, which can be removed by the plastic tab which can be pulled back one stretching it side way.

Viva Questions:

- 1) Define assembling of a system?
- 2) Explain the steps involved in the installation of the mother board?
- 3) What is the use of pin 1 indicated on the processor?
- 4) What is the use of locking level at the processor slot?
- 5) What is the steps used in removing the drive ?
- 6) What is DIMM?

Lab3: Assembling Laptop Computer

THEORY;-Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

AIM: To assemble and disassemble the system

Why should one learn about hardware?

- 1.Troubleshoot you and save time.
- 2.Knowing about system internals and components.
- 3.Very easy installation for modern hardware.
- 4.Install extra memory.
- 5.Removing components.

Safety Precautions:

- 1.Beware of electrostatic discharge (ESO)
- 2.Build computer on a hard surface, away from concepts.
- 3.Wear shoes and the shortsleeved cotton wear.
- 4.Use Phillips, head screw driver.
- 5.Keep the components away from moisture.
- 6.Avoid using pressure while installing.

Material Required

- Laptop

- Mini screwdriver
- Anti-static wrist strap
- Magnifying glass

Procedure for assembly

Step 1: Keep track of screws

Step 2: Installation of processor

Step 3: Pop in the video card

Step 4: Set up the drive

Step 5: Add memory

Step 6: Final assembly

Disassembly of Laptop

Step 1: Removal of battery

Step 2: Separation of palm rest

Step 3: Removal of bottom case

Step 4: Disconnecting the hard drive cable from the motherboard

Step 5: Remove the hard drive

Step 6: Removal of RAM

Step 7: Disconnection of cable

Step 8: Removal of USB LED status board

Step 9: Removal of screw fixing the optical CD/ DVD drive connector board

Step 10: Removal of the cooling fan

Step 11: Removal of DC power jack

Step 15: Removal of screw securing the board

Step 12: Removal of motherboard

Laptop LCD Screen Removal

Procedure

It is possible to disassemble the LCD screen of a laptop independently. Previous steps are not required for disassembling the display, you only need to disconnect the battery before you start.

Step 1: Separate the display bezel from the back cover. You will have to wiggle the bezel to unfasten it from the cover. Start on the top and move to the sides for the display assembly.

Step 2: On the bottom, the bezel is attached to the LCD screen with adhesive tape. Carefully separate it from the screen.

Step 3: Remove the bezel completely.

Step 4: Remove the four screws securing the LCD screen to the side brackets

Step 5: Separate LCD screen from the back cover and place it the front side down on the keyboard.

Step 6: Now you can access the video cable connector

Step 7: Peel off the grounding tape from the screen.

Step 8: Peel off the clear tape securing the connection and unplug the video cable from the screen

Step 9: Remove the LCD screen completely and replace if necessary

Step 10: You can find a new LCD screen using the model number from the original one.

LAB 4: Laptop/Desktop OS Installation

THEORY:- Every student should individually install MS windows OS on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

INSTALLATION OF WINDOWS OS:

AIM: To install Windows 7 Operating System

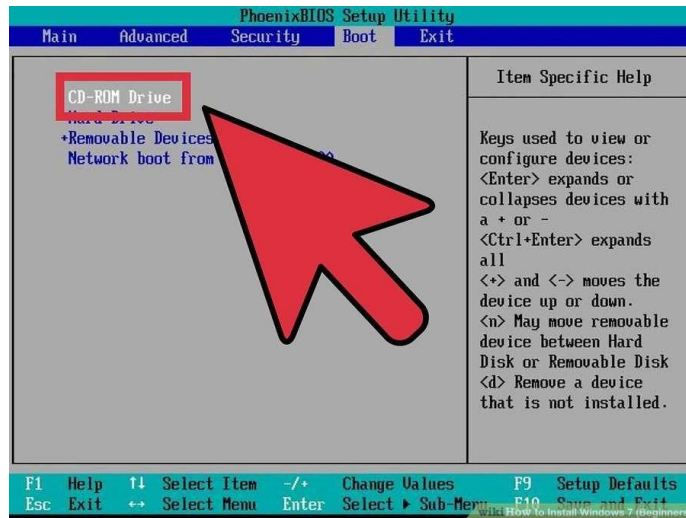
BASIC FILE SYSTEMS:

- FAT: File Allocation Table.
- NTFS: New Technology File System. Format and create using NTFS partition

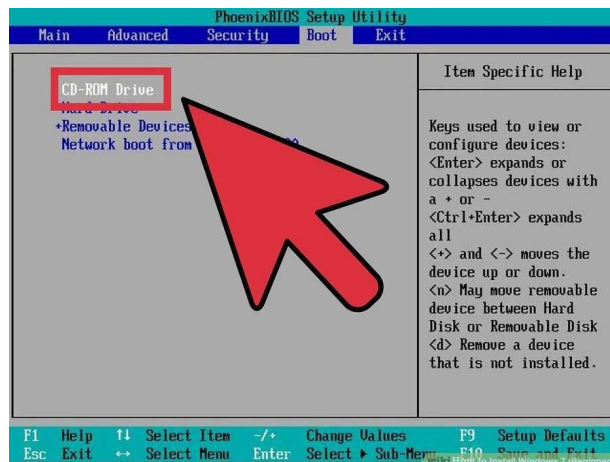
BASIC STEPS IN INSTALLATION: -

A clean install is intended for users who want to freshly install Windows on their computer (by deleting all of the data on the hard disk and then installing Windows) or computers that do not have an operating system yet.

1. Enter your computer's BIOS. Turn off the computer that you want to install Windows on then turn it back on. When the BIOS screen appears or you are prompted to do so, press Del, Esc, F2, F10, or (depending on your computer's motherboard) to enter the system BIOS. The key to enter the BIOS is usually shown on the screen.

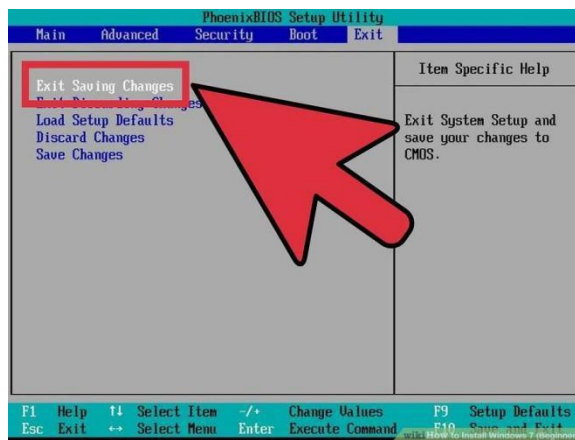


2. Find your BIOS's boot options menu. The boot options menu of your BIOS may vary in location or name from the illustration, but you may eventually find it if you search around. If you can't find the boot options menu, search the name of your BIOS (most likely located in the BIOS menu) online for help.

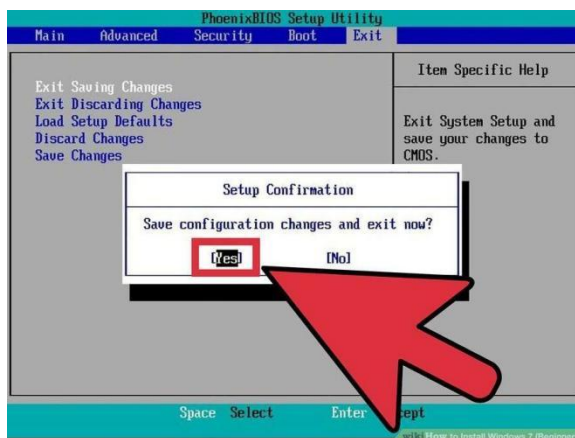


3 Select the CD-ROM drive as the first boot device of your computer.

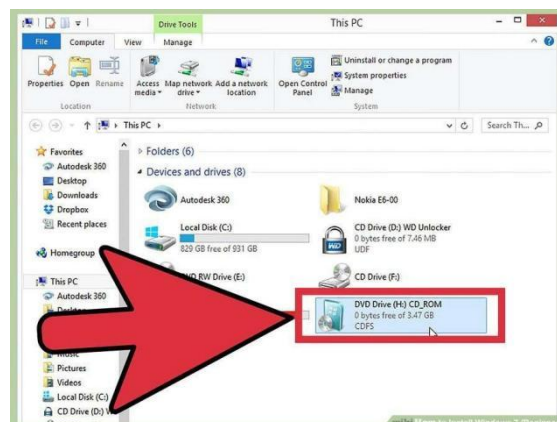
Although this method may vary among computers, the boot options menu is typically a menu of movable device names where you should set your CD-ROM drive as the first boot device. It can also be a list of devices that you can set the order of their boot on. Consult a manual or the internet for help if you're stuck.



4 Save the changes of the settings. Press the button indicated on the screen or select the save option from the BIOS menu to save your configuration.



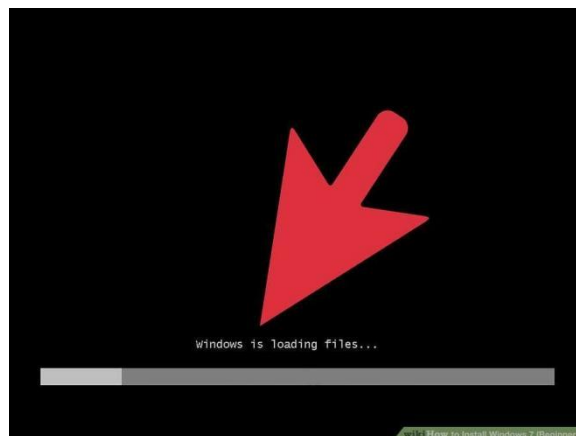
5. Shut off your computer. Either turn off the computer by choosing the shut-down option in your current operating system, or hold the power button until the computer powers off



5. Power on the PC and the insert the Windows 7 disc into your CD/DVD drive

6. Start your computer from the disc. After you have placed the disc into the disc drive, start your computer. When the computer starts, press a key if you are asked if you would like to boot from the disc by pressing any key. After you choose to start from the disc, Windows Setup will begin loading.

If you are not asked to boot from the disc, you may have done something wrong. Retry the previous steps to solve the problem.



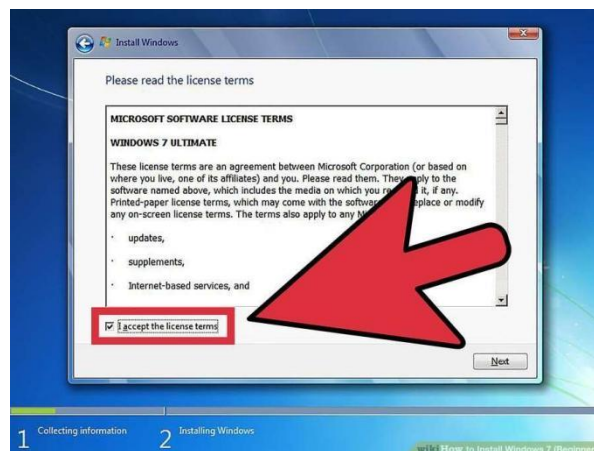
7. Choose your Windows Setup options. Once Windows Setup loads, you'll be presented with a window. Select your preferred language, keyboard type, and time/currency format, then click *Next*.



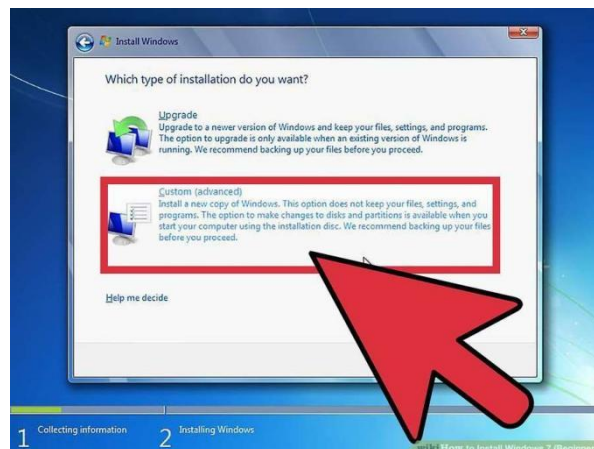


8. Click the ***Install Now*** button.

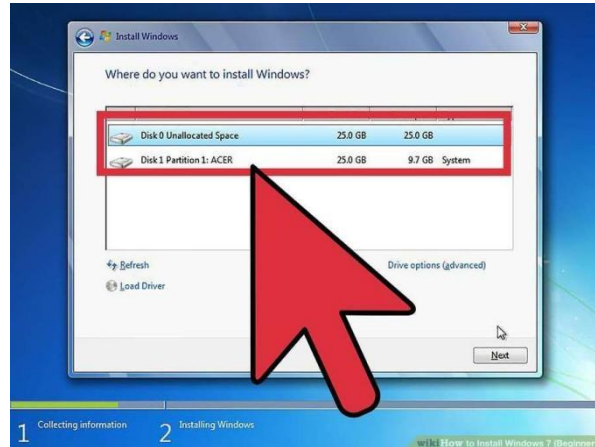
9. **Accept the License Terms.** Read over the Microsoft Software License Terms, check *I accept the license terms*, and click *Next*.



10. Select the ***Custom*** installation.



11. Decide on which hard drive and partition you want to install Windows on. A hard drive is a physical part of your computer that stores data, and partitions "divide" hard drives into separate parts.



If the hard drive has data on it, delete the data off of it, or format it.

Select the hard drive from the list of hard drives.

Click Drive options (advanced).

Click Format from Drive options.

If your computer doesn't have any partitions yet, create one to install Windows on it.

Select the hard drive from the list of hard drives.

Click Drive options (advanced).

Select New from Drive options.

Select the size, and click OK.

12. Install Windows on your preferred hard drive and partition. Once you've decided on where to install Windows, select it and click *Next*. Windows will begin installing.



Viva Questions:

- 1) NTFS stands for ?
- 2) What is the use of product key in the installation process of a software?
- 3) Installing - What does "upgrade" mean?
- 4) Write the procedures to create bootable disc?
- 5) Describe different kinds of Microsoft Operating systems?
- 6) What are the pre-arrangements for installing the windows OS?
- 7) What is virtual memo

Installation of Operating Systems –LINUX

Every student should install Linux on the computer. This computer should have windows installed. The system should be configured as dual boot with both windows and Linux. Lab instructor should verify the installation and follow it up with a viva.

AIM: To install Linux operating system(Ubuntu 12.04 LT)

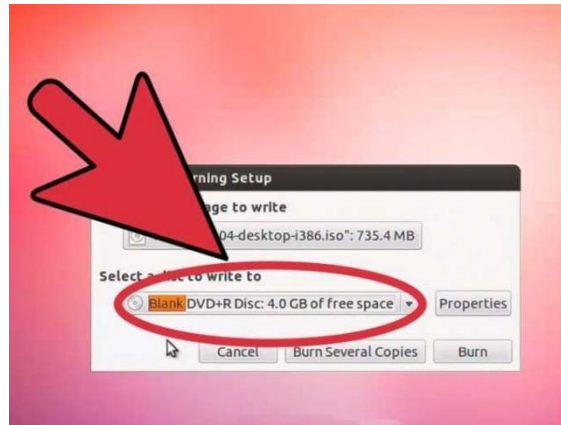
PROCEDURE:

Ubuntu is one of the most popular and easy-to-use versions of Linux available, and you can download and install it absolutely free. All you need is a CD burner and an internet connection, and you can have Ubuntu up and running in just a few minutes.

- 1. Download the Ubuntu image.** Ubuntu is available for free from the Ubuntu website. It is downloaded in ISO format, which needs to be burned to a CD or DVD before you can use it. Most newer computers (built after 2011) will use the 64-bit version, while older computers will need the 32-bit version.



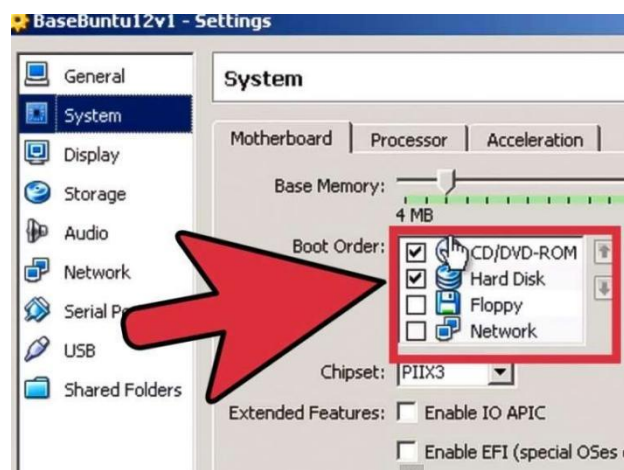
- 2. Burn the image to a disc.** There are a variety of free image burning programs available, and Windows 7, 8 and Mac OS X all have image burning capabilities built-in.



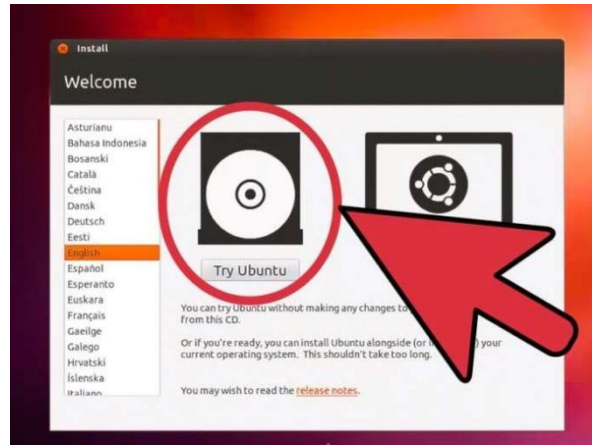
- In Windows 7 and 8, navigate to the downloaded ISO file and double-click on it. Insert a blank disc into your burner, and follow the on-screen instructions to burn the disc
- In Mac OS X, open the Disk Utility. This is located in the Utilities folder, in your Applications folder. Insert your blank disc into your disc drive. Drag and drop the ISO file into the left frame of Disk Utility. Select the ISO file in the frame and click Burn.

3. Set your computer to boot from the CD/DVD drive. In order to install Ubuntu, you will need to set your computer to boot from discs. This is because the installation has to occur before your existing operating system is loaded from the hard disk.

- When you reboot your computer, press the BIOS setup key to enter your BIOS menu. In the Boot section, select your CD/DVD drive as the primary boot device. Once you have it set, save and exit the BIOS. Your computer will reset again.
- In Windows 8, hold the Shift key while clicking Restart in the Power menu. This will reboot the computer into the Advanced Startup Options. From here, you can choose to boot from a CD or DVD.

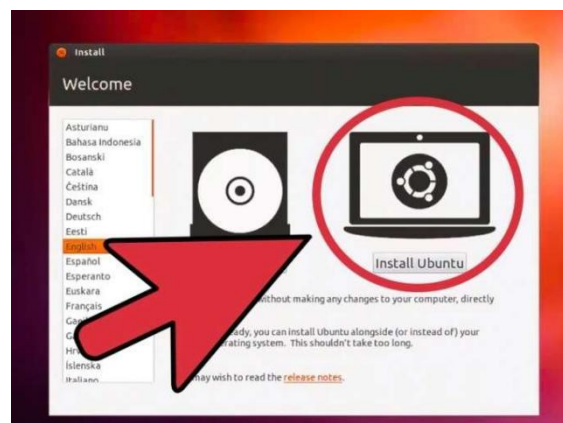


4. Try Ubuntu before you install. If you'd like to take Ubuntu for a test drive before you commit to installing it, you can run Ubuntu directly from the installation CD without making any changes to your computer. Click the "Try Ubuntu" button to test it out.



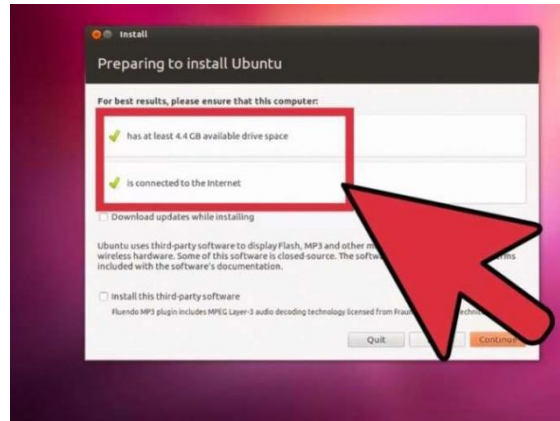
5. Start the installation process. Click the Install Ubuntu button to begin the installation. If you are trying out Ubuntu before installing, you can start the installation by opening the Install file on the test desktop.

- You will need around 5 GB of free space on your computer to install Ubuntu.
- Installation works best if you are connected to the internet. The easiest way to do this is to plug an Ethernet cable directly into your computer from the router.
- You can also connect to a wireless network if Ubuntu recognizes your wireless adapter.



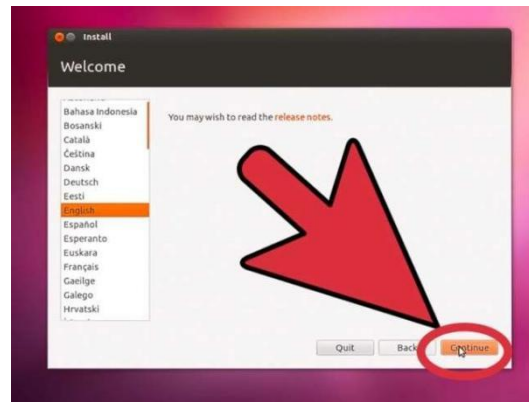
6. Check the boxes in the “Preparing to install Ubuntu” window.

Ubuntu can automatically download updates during the installation, and can install software that will allow you to play MP3 files and Flash video (YouTube). Installing these updates and software during installation will save you a significant amount of time and headaches when the operating system is installed.

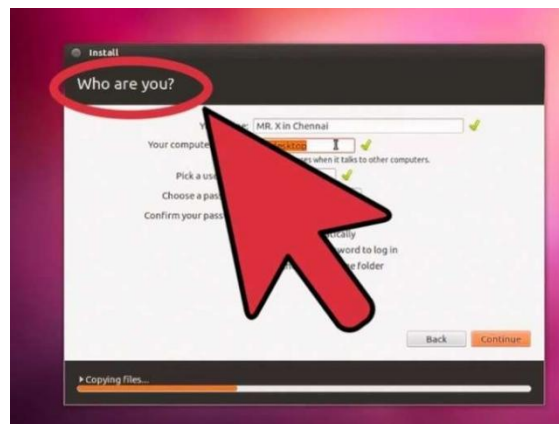


1.Choose your installation type. You can choose to install Ubuntu alongside your existing operating system, you can replace your existing operating system, or you can create new partitions yourself. If you choose to install alongside your existing operating system, you will be given the option to set how much space is allocated for each operating system.

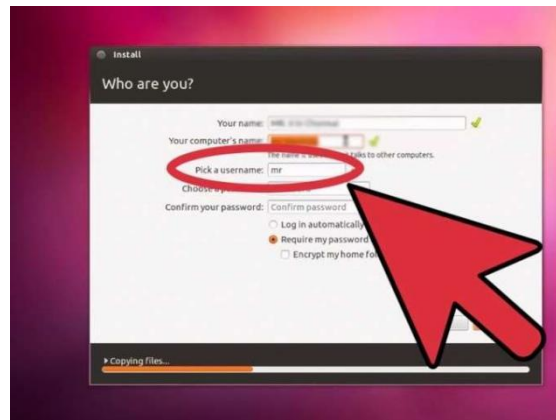
- If your computer doesn't have an operating system installed yet, you will need to select "Something else" and create a partition to install Ubuntu on. Make sure that the partition is formatted as Ext4.
- If you install Ubuntu alongside your existing operating system, you will be given the option to choose your operating system every time your computer is rebooted.
- If you replace your existing operating system with Ubuntu, you will lose all of your files and programs. Make sure that you have everything backed up that you want to save.



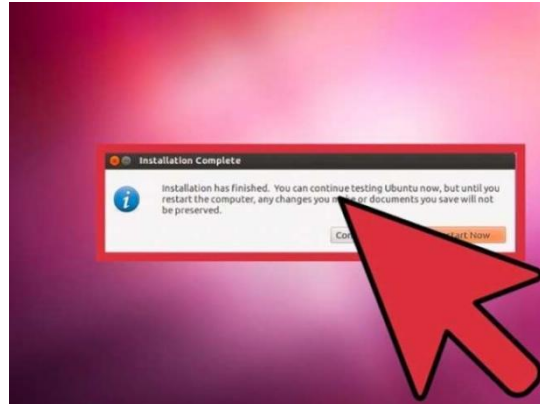
2. Select your user options. Ubuntu will ask for your location in order to set the correct timezone. If you are connected to the internet, this should happen automatically. You will also need to select your keyboard layout, which is typically automatically picked for you.



1. Select a username and password. In the “Who are you?” window, you will need to enter your name, your computer’s name, and a username and password that you will use to log in. You will need the password when making administrative changes to the computer as well.



2. Wait for the installation to complete. Once you have set your options, Ubuntu will begin to install. This can take 30 minutes or more depending on your computer’s speed. Tips for using Ubuntu will be displayed above the progress bar. Once the installation is complete, you will be prompted to reboot your computer. Ubuntu is ready to use.



- If you have multiple operating systems installed, you will be given the option to choose the one you want to load when your computer starts.
- If you don't have any other operating systems installed, Ubuntu will begin automatically.

Viva Questions:

- 1) Give the advantages of Linux over other OS?
- 2) What do you mean by open source OS?
- 3) What do you mean by dual boot systems?
- 4) Where the all directories are stored in Linux
- 5) Which command is used to make directory
- 6) Which command is used to change directory
- 7) Which command is used to copy from one file to another file?
- 8) Which command is used to move data from one file to another file?

Lab 5: Networking and Internet Setup

THEORY:-Here every student should know how to get connected to Local Area Network and access the Internet. In the process they configure the TCP/IP setting. They should know how to access the websites and email. If there are no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN

AIM: To learn Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally, students should demonstrate, to the instructor, how to access the websites and email

THEORY: The internet is a worldwide, publicly network of interconnected computer networks

Network Cables

Several standard types of **network cables** exist, each designed for specific purposes.

Coaxial Cables

A Coaxial cable is a cable used in the transmission of video, communications, and audio. This cable has high bandwidths and greater transmission capacity. Most users relate to a coaxial or coax cable as a cable used to connect their TVs to a cable TV service. However, these cables are also used in networks and what allow a broadband cable Internet connection using a cable

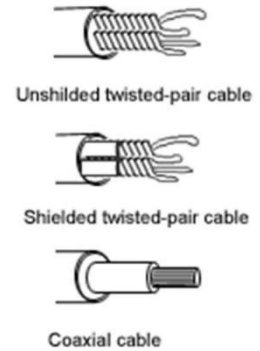
modem. The picture is an example of a coaxial cable. As can be seen in this picture, the cable is a thick cable with a metal male connector end that is screwed onto a female connector.



Twisted-pair cable

A twisted-pair cable is a cable made by inter twinning two separate insulated wires. There are two twisted pair types: shielded and unshielded. A Shielded Twisted Pair (STP) has a fine wire mesh surrounding the wires to protect the transmission an Unshielded Twisted Pair (UTP) do not. Shielded cable is used in older telephone networks, as well as network and data communications to reduce outside interference. The illustration gives an example of how the inside of these looks.

Networking Cables



Crossover Cable

Networking cable that connects two computers or network devices directly to one another. When purchasing this cable, the packaging must indicate that it's a crossover cable for the required network interface; otherwise, it's likely a typical straight through cable. Often, crossover cables are used to connect two computers with network cards together without using a network hub, network router, or network switch.

CAT5

Alternatively referred to as an Ethernet cable, a CAT5 or Category 5 is a description of network cabling that consists of four twisted pairs of copper wire terminated by an RJ-45 connector. Cat5 has a maximum length of 100m, exceeding this length without the aid of bridge or other network device could

cause network issues. The picture shows an example of what a CAT5 cable may look like with the connector, as well as the other end of the cable with each of the wires it contains.



Fiber Optic Cable

A fiber optic cable defined in IEEE 802.8 is cable that contains optical fibers (usually glass) coated in plastic that are used to send data by pulses of light. The coating helps protect the fibers from heat, electromagnetic interference from other types of wiring, as well as some protection from ultraviolet rays from the sun. Fiber optics allow for a much faster data transmission than standard copper wires, because they have a much higher bandwidth. They are common amongst corporate networks or world-wide networks, such as Internet backbones, because of the capabilities of the cable.

NETWORKING DEVICES:

Hub

When referring to a network, a hub is the most basic networking device that connects multiple computers or other network devices together. Unlike a network switch or router, a network hub has no routing tables or intelligence on where to send information and broadcasts all network data across each connection. Most hubs can detect basic network errors such as collisions, but having all information broadcast to multiple ports can be a security risk and cause bottlenecks. In the past, network hubs were popular because they were cheaper than a switch or router. Today, switches do not cost much more than a hub and are a much better solution for any network.

Switch

A switch is a piece of a physical circuitry component that governs the signal flow. Having a switch or toggle switch allows a connection to be opened or closed. When opened, the switch allows a signal or power to flow through the connection. When closed, the switch stops the flow and breaks the circuit connection.

On a network, a switch is a hardware device that filters and forwards network packets, but often not capable of much more. A network switch is more advanced than a hub but not as advanced as a router.

NETGEAR 5 Port Network Switch

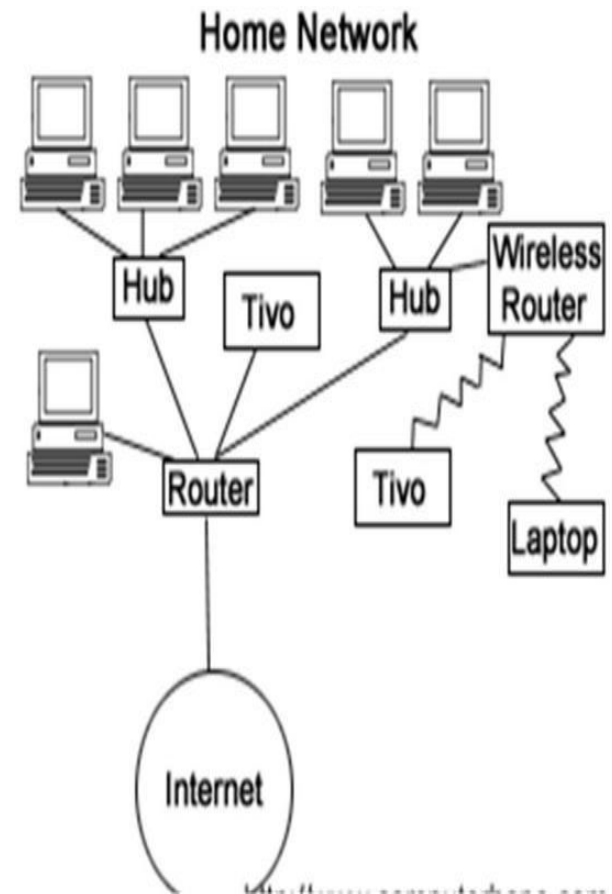


Router

A router is hardware device designed to receive, analyze and move incoming packets to another network. It may also be used to convert the packets to another network interface, drop them, and perform other actions relating to a network.

A router has a lot more capabilities than other network devices, such as a hub or a switch that are only able to perform basic network functions. For example, a hub is often used to transfer data between computers or network devices, but does not analyze or do anything with the data it is transferring. By contrast routers can analyze the data being sent over a network, change how it is packaged, and send it to another network or over a different network. For example, routers are commonly used in home networks to share a single Internet connection between multiple computers.

In the above example, of a home network, there are two different types of a router: the router and the wireless router. In this example, the router allows all the computers and other network devices to access the Internet. The wireless router allows a laptop to wirelessly connect to the home network and access the Internet as well. Below are some additional examples of different types of routers used in a large network.



- **TCP/IP(Transmission Control Protocol/Internet Protocol):** Collection of methods used to connect servers on the internet and to exchange data.
- **HTML(Hyper Text Markup Language):** The coding used to control the look of documents on the web
- **HTTP(Hyper Text Transfer Protocol):** Part of a url that identifies the

location as one that uses HTML

- **IP(Internet Protocol):** A format for contents and addresses of packets of information sent over the internet
- **IP ADDRESS:** An identifier for a computer or device on a TCP/IP network

PROCEDURE:

- Goto start>control Panel
- open Network Connections
- Click create a new connection and then click next
- The new connection wizard window opens ,click next to continue
- Choose one of the options in the next dialog box
- Choose one of the three options in the next dialog box
 - ✓ If you do not have an internet account click choose from a list of ISPs and then click next
 - ✓ If you have an account click Set up my connection manually
 - ✓ If you have a CD from the ISP click use the CD I got from an ISP and then click next
- Follow the next steps as per the option you selected.

Viva Questions:

1. What is subnet mask?
2. What do you mean by network?
3. Briefly describe NAT.
4. What is NIC?

Lab 6: SOFTWARE INSTALLATION

Configuring Firewalls and installation of Antivirus software

Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to first install an antivirus software, configure their personal firewall and windows update on their computer. Then they need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms

PURPOSE: To learn various threats on the internet and configure the computer to be safe on the internet.

THEORY: Antivirus:

Antivirus software is a program that either comes installed on your computer or that you purchase and install yourself. It protects your computer against most viruses, worms, Trojan horses and other unwanted invaders that can make your computer sick.

Firewall:

A firewall is a special software or hardware designed to protect a private computer network from unauthorized access. A firewall is a set of related programs located at a network gateway server which protects the resources of the private network from users from other networks.

Software firewalls

Software firewalls are designed to protect a computer by blocking certain programs from sending and receiving information from a local network or the Internet. The image to the right shows the icon for Windows Firewall, an example of a firewall software program.



Hardware firewalls

Hardware firewalls are found on most network routers and can be configured through the router setup screen. The image shows the ZyXEL ZyWALL, which is a Unified Security Gateway with a firewall and other security features.

ZyXEL ZyWALL with Firewall



PROCEDURE:

Installing Symantec antivirus for Windows:

- Insert Symantec antivirus CD into your CD drive
- Double click on the Symantec-setup.exe
- The installer will open
- Click **next** to proceed
- License agreement will open .Click **I accept the terms of the license agreement** and then click **next**.
- Follow the instruction on the screen to complete the installation.

Get Computer Updates:

- Click **start> settings>control panel**
- Click **Automatic Updates** icon to open Automatic Updates dialog box
- Check the box **Keep my computer up to date**
- Choose a setting
- Click OK

Block Pop ups:

- In the IE open **tools>pop-up blocker**
- Click on **Turn on Pop- up**

blockerWindows Firewall:

- Go to **Start>control panel>Network and InternetConnections>windows firewall**
- In the general tab check the **On(recommended)** box
- If you don't want any exceptions check on **Don't allow exceptions box**

Viva Questions:

- 1) What is antivirus software?
- 2) Define virus
- 3) Define worm
- 4) What are the advantages of antivirus software?
- 5) What are the types of antivirus software's available?

Lab 7: Server Installation and Backup

THEORY:-

A server is a computer program or device that provides a service to another computer program and its user, also known as the client. In a data center, the physical computer that a server program runs on is also frequently referred to as a server. That machine might be a dedicated server or it might be used for other purposes.

AIM:- To understand about the various types of server and its installation process.

TYPES OF SERVER:-

- Web server: a computer program that serves requested HTML pages or files. In this case, a web browser acts as the client.
- Application server: a program in a computer in a distributed network that provides the business logic for an application program.
- Proxy server: software that acts as an intermediary between an endpoint device, such as a computer, and another server from which a user or client is requesting a service.
- Mail server: an application that receives incoming emails from local users -- people within the same domain -- and remote senders and forwards outgoing emails for delivery.
- Virtual server: a program running on a shared server that is configured in such a way that it seems to each user that they have complete control of a server.
- Blade server: a server chassis housing multiple thin, modular electronic circuit boards, known as server blades. Each blade is a server in its own right, often dedicated to a single application
- File server: a computer responsible for the central storage and management of data files so that other computers on the same network can access them.
- Policy server: a security component of a policy-based network that provides authorization services and facilitates tracking and control of files.
- Database server: this server is responsible for hosting one or more databases. Client applications perform database queries that retrieve data from or write data to the database that is hosted on the server.
- Print server: this server provides users with access to one or more network-attached printers -- or print devices as some server vendors call them. The print server acts as a queue for the print

jobs that users submit. Some print servers can prioritize the jobs in the print queue based on the job type or on who submitted the print job.

How to Install FTP Server

Before discussing about Installing FTP Server let's have a look first on what is server. Server is a centralized unit (whether a PC or a server machine) in a network system for providing and maintaining the internet service to the number of users attached with it. Server is system act as a host which is responsible to provide services to the clients and also take their command.

Internet Information Services (IIS) is an application to support web browsing activities. It supports following protocols; FTP, FTPS, SMTP, NNTP, HTTP. File Transfer Protocol (FTP) performs two major functions. With the help of FTP one can get better source for virus detection, access free software program by downloading them from remote point. It also helps to secure the data by uploading them on a remote place. File transferring protocol when install in PC act as an FTP server.

Now understanding all the basic units what are they? Why we need it? Let see how can we install FTP Server?

Procedure: Step – 1

1. First click the start button and go to control panel.
2. Now select Add / Remove Windows components, you will see Windows Components Wizard.
3. In components not check rather click the Application Servers, Internet Information Services (IIS) and see the details.
4. By clicking the details you will see Common Files, File Transfer Protocol (FTP) Service and Internet Information Service (IIS) Manager now check their boxes.
5. You can check other programs related to Internet service you need and then click OK.
6. After clicking OK, a prompt "Next" will come in front of you asking for the installation command.
7. Locate the file from the program or insert CD.
8. After the program is installed click Finish.

After the installation of IIS is complete now configure the FTP service before you go for the application.

Step – 2

1. Run Internet Information Service Manager
2. You will see server _ name, presently expand it.
3. FTP Sites are displayed, select default FTP Site
4. Open FTP'S Site properties.
5. Go for the Security Accounts.
6. In the security account tab click the Allow Anonymous Connections.
7. Select Allow Only Anonymous Connections. At this point you will configure only anonymous connections.
8. Open the Home Directory tab.
9. Look for the Read and Log in check boxes and mark them if not marked already.
10. Look for the Write check box and clear it if it is marked.
11. Now click OK. You have done the configuration.
12. Close all the open windows and quits IIS Manager.

Now your system is equipped with the FTP Server to perform the relevant functions. Place the data by copying or moving the files which you need to make available to others.

Viva Questions:

- 1) What is server?
- 2) Explain different types of server?

Lab 8: Hardware and Software Troubleshooting

It is a process of sorting out the problems.

The set of steps we follow during trouble shooting are:

- a) Identify the problem.
- b) Identify the set of solutions.
- c) Analyze any one solution which is less costly and more efficient.
- d) Apply the solution.
- e) Test the output.

Trouble shooting is generally of 2 types:

- 1) Hardware troubleshooting: deals with the process of eliminating bugs related to PC components.
- 2) Software trouble shooting: refers to removal of problems rising due to system software or application software.

HARDWARE TROUBLESHOOTING:

The list of techniques under this are:

- Hard disk troubleshooting
- Monitor troubleshooting
- Modem troubleshooting
- Keyboard troubleshooting.

a) HARD DISK TROUBLESHOOTING

It contains platters, head arms, classes and head actuators. Most of the HDDS are connected by either the ATA(i.e.; another name for IDE) or we use SCSI(small computer system interphase) cables.

The difference between SCSI and IDE is

- Using IDE we can connect almost two hard disks
- Using SCSI we can connect 7to 15 hard disks

- Maintenance of SCSI is very difficult when compared to IDE
- The cost of SCSI is more compared to IDE.

Problems related to Hard disk

- ✓ If the HDD is not detected inside the configurations
- ✓ If the system files on the hard disk drive are missing or if they are corrupt.

The computer does not boot up when turned on

Solutions related to hard disk

- ✓ Check whether HDD is physically available or NOT.
- ✓ Check if the cable is connected to the hard disk and if it is detected.
- ✓ On the existing hard disk make sure that you scan the disk very well using the current version of antivirus
- ✓ Make sure that you are setting up a disk or changing its partitions.

b) MONITOR TROUBLESHOOTING:

Which is physically connected to video cards like TV monitor also contains CRT (cathode ray tube). CRT contains 3 guns, red, green and blue. These guns stream electrons from left to right.

Set of troubleshooting etc:

PROBLEM: If the picture is not visible

SOLUTION: Check the signal cable is into the power socket or not.

Set of troubleshooting steps are: Check if the signal cable is connected to graphic adapter.

Check brightness of the screen.

PROBLEM: If the screen is too bright or dark.

SOLUTION: Check the brightness value. Check if the specified voltage is applied or not. Check if the frequency is horizontal as well as

vertical.

PROBLEM: If the screen is blurred

SOLUTION: Move all the objects that emit magnetic field away. Check whether the specified is applied or not. Check if the signal timing of system is within the specification.

MONITOR ICONS:

- ✓ Power
- ✓ Brightness
- ✓ Contrast
- ✓ Horizontal size
- ✓ Vertical size
- ✓ Full screen
- ✓ Monitor status
- ✓ Language

c) MODEM TROUBLESHOOTING:

We can call a modem as translator, demodulator, which describes a method by which digital data is converted to analog while receiving the data by the computer it is converted to digital from analog.

There are 2 types of modem which are available they are RJ11 and RJ45. RJ11 consists of 6 pins and RJ45 of 8 pins.

Steps of troubleshooting for modem are:

- ✓ Check the telephone cable connections 10 feet or less than contain a RJ11 cable connection.
- ✓ Check to ensure that you are plugged to the correct modem jack i.e; RJ45 will not fit in RJ11 but RJ11 will fit into RJ45.
- ✓ Check for the error message i.e wrong user name and password the contact your ISP (Internet service provider) for the correct account settings.
- ✓ Reduce your port speed in your modem settings.

- ✓ Check the initialization settings
- ✓ Use hyper terminal on another tool to test the modem.

d) KEYBOARD TROUBLESHOOTING:

- ✓ If the keyboard is not working then check if the keyboard cable is inserted properly in the connector on the motherboard.
- ✓ Swap a non-functional keyboard with a functioning one.

Other Hardware Issues:

1. Disk Boot Failure
2. S.M.A.R.T Command Failed
3. Automatic Restart of your Computer
4. Fix MBR issues
5. Hide computer Hard disk Drive
6. How to Change Computer Hard Disk Drive Letters
7. Protect your PC from Over heating
8. Use Flash Memory (USB Memory) as Virtual RAM
9. Problems in booting computer in safe mode
10. Connect two Systems using USB cable
11. Bios Backdoor Passwords for Laptops
12. Reset Bios Passwords

Software Troubleshooting

Students have to be given a malfunctioning CPU due to system software problems. They should identify the problem and fix it to get the computer back to working condition. The work done should be verified by the instructor and followed up by the viva.

AIM: Software troubleshooting

PROCEDURE:

Error messages encountered during boot before Windows loads

Ensure that your computer BIOS settings are correctly configured to the hardware that is installed in your computer

1. Issues and questions related to the Windows taskbar.

2. Information about the Taskbar.

Verify Auto hide is not activated by following the below steps

1. Press CTRL + ESC or your Windows key to make Start menu appear
2. Click Settings and choose Taskbar & Start menu
3. Verify that Auto Hide is not checked

Shutdown, restart, logoff, and/or poweroff the computer

Users can shutdown Windows through the command line using rundll32 and the shell32.dll file.

Windows restarts without warning

This issue could be caused by any of the below possibilities

1. Software issue or error
2. Hardware issue or error
3. Heat related issue.
4. Computer virus.
5. Issue with operating system.

Error messages while windows loading

1. If you have recently installed or changed something that could have caused normal windows to stop loading, try loading the last known good configuration
2. If you are unable to get into Normal windows and believe that removing or uninstalling a program or changing a setting may help

enable you to get into windows, boot the computer into Windows OS safe mode

3. If your computer has worked fine in the past but recently has been experiencing the issue you are encountering run the system restore option to restore the computer to an earlier date

Other error messages that occur while windows is loading or after windows is loaded

1. If error occurs but windows still loads, verify no issues or conflict exists in device manager
2. Ensure that if programs are loading automatically that these errors are not associated with these programs
3. Make sure Windows OS is up to date by checking Microsoft windows update page
4. If your computer has virus protection installed make sure that it is up to date and that no virus are being detected
5. If your computer has worked fine in the past but recently has been experiencing the issue you are encountering run the system restore option to restore the computer to an earlier date

Viva Questions:

- 1) Define trouble shooting?
- 2) Mention a few problems associated with Modem.
- 3) What is the function of a modem?
- 4) What are the different key board problems?
- 5) Give the preventive measures for better computer utilization?
- 6) Define a software What are the general software problems we encounter?

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